

Design and Simulation of Light Weight High Gain Horn Antenna at Ku-Band using HFSS

Review 2: Baseline Design and Initial Simulation

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Outline

- 1 Antenna Design Methodology
- 2 HFSS 3D Geometric Modeling
- 3 HFSS Simulation Setup
- 4 Initial Simulation Results
- 5 Literature Expansion
- 6 Review 3 Roadmap

Antenna Dimensions and Theoretical Calculations

Target Specifications: Operating Frequency (f_c) = 15 GHz, Minimum Gain = 15 dBi.

1. Waveguide Feed (Standard WR-62):

- Supports dominant TE_{10} mode in Ku-band.
- Inner Width (a) = 15.799 mm, Inner Height (b) = 7.899 mm.

2. Optimum Horn Aperture Calculations ($G_{lin} \approx 31.62$):

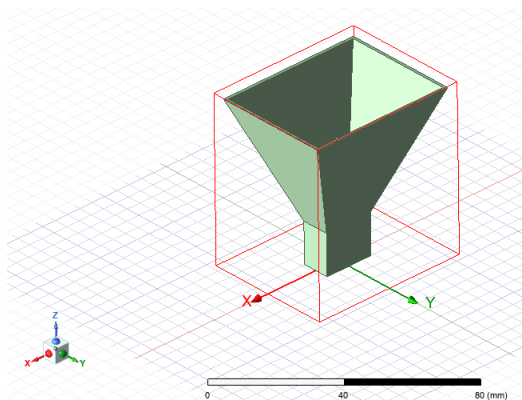
- H-Plane Width (A) $\approx 0.45\lambda\sqrt{G_{lin}} = 50.6$ mm
- E-Plane Height (B) $\approx 0.35\lambda\sqrt{G_{lin}} = 39.3$ mm

3. Axial Flare Length:

- Length (L) $\approx \frac{A^2}{3\lambda} = 42.6$ mm

Geometry Construction (Physical Shell):

- Instead of simulating a simple vacuum block, a physical antenna shell was modeled to represent a 3D-printable structure.
- The inner and outer profiles of the WR-62 feed and pyramidal flare were drafted using coordinate boxes and polylines.
- The *Subtract* Boolean operation was utilized to hollow out the inner volume, giving the antenna walls a defined physical thickness.
- The structural material was explicitly assigned as **ABS Plastic**, laying the groundwork for lightweighting.



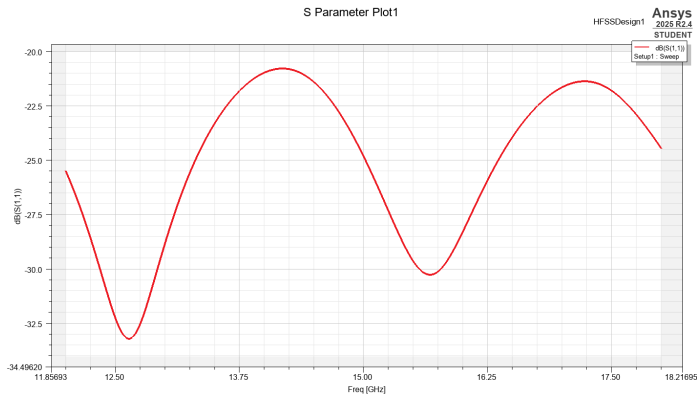
Simulation Setup and Boundary Conditions

- **Solution Type:** Driven Modal (required for proper TE_{10} wave propagation through a single-conductor waveguide).
- **Boundary Conditions:** Appropriate conductive boundaries were assigned to the ABS shell surfaces to facilitate electromagnetic reflection and propagation.
- **Excitation:** A **Wave Port** was assigned to a 2D rectangular sheet flush against the rear face of the WR-62 feed to inject the RF signal.
- **Radiation Box:** The *CreateRegion* function was used to automatically generate a vacuum airbox with a 2 mm absolute offset padding around the antenna to compute far-field parameters.

Impedance Matching: Return Loss (S_{11})

Return Loss Analysis:

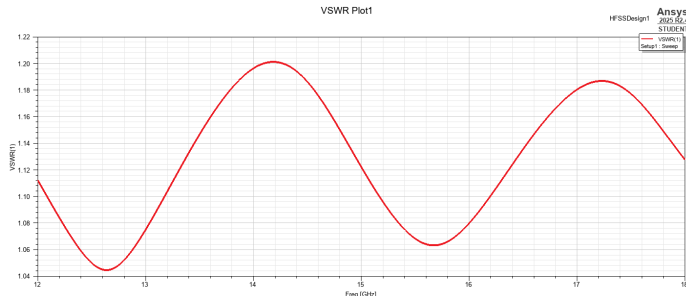
- Frequency sweep conducted across the Ku-band (12–18 GHz).
- The antenna exhibits excellent wideband impedance matching, staying well below the -10 dB operational threshold.
- At our exact 15 GHz target, the S_{11} operates at approximately **-25 dB**, with a maximum resonance dip reaching **-33 dB** near 12.6 GHz.



Impedance Matching: VSWR

VSWR Performance:

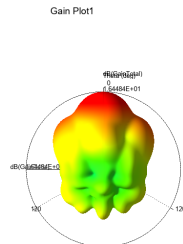
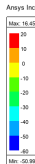
- The Voltage Standing Wave Ratio validates highly efficient power transmission.
- The linear VSWR peaks at only **1.2** across the entire simulated spectrum, remaining comfortably below the strict ≤ 2.0 aerospace standard.
- At 15 GHz, the VSWR is approximately **1.11**, indicating nearly zero signal reflection back into the feed.



Radiation Pattern and Gain

Far-Field Characteristics:

- 3D Polar plot generated at the 15 GHz center frequency.
- The model achieved a Peak Realized Gain of **16.45 dBi**.
- This successfully exceeds our baseline project requirement of 15 dBi, proving the ABS shell geometry is highly effective.
- *[Vinay's text analyzing Half-Power Beamwidth (HPBW) will go here]*



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Updated Literature Survey (Monish & Vinay)

[Placeholder: Monish and Vinay's new literature survey table will be inserted here]

Targeting Weight Reduction:

- *[Placeholder: Rammanish's text on investigating lightweight polymers]*
- *[Placeholder: Rammanish's text on 3D printing and electroless plating techniques]*
- *[Placeholder: Planned structural modifications in HFSS to reduce mass while maintaining >15 dBi gain]*